

Soumendu Nandi

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Data and Generative AI Product Engineering and Architecture (Python, .Net), Azure Cloud Adoption, Azure DevOps

Professional Summary

With over 17 years of experience as a Solution Architect and Software Product Engineer, I have designed and delivered cloud-native enterprise applications. My expertise includes Object Oriented Programming using Python, C#, Azure Cloud, clean coding and code review, DevOps automation, Application Architecture, Cloud Architecture. I have a proven track record of reducing time-to-market and ensuring reliability in regulated domains such as Life Science, Pharma, Banking, Transport, and logistics. I am passionate about Generative AI Application development, digital transformation, AI adoption, cloud adoption, reliability, and cost optimisation.

Core Competencies

As a Software Architecture Expert, I possess a strong understanding of software architecture principles, including scalability, clean coding, clean architecture, modularity, and security. I am proficient in delivering solutions using cloud computing platforms, including private clouds and distributed computing environments.

I collaborate closely with the platform team, Enterprise Architects, and Gen AI Innovation Leadership to drive business solutions for controlled and audited environments. My focus is on enhancing reusability, Solution Governance, accountability, Architecture Governance, Reusable and shared Integration Patterns, and total cost of ownership (TCO).

I align high-level design for data acquisition from upstream applications with the organisation's strategy, FinOps, and Data Governance. I lead technical teams and effectively communicate with non-technical stakeholders, emphasising attention to detail, problem-solving, and solution structuring for Generative AI solutions for enterprise.

Engaged into AI accountability, LLM fine-tuning, and LLM evaluation within the product development pipeline.

I have a understanding of artificial intelligence and machine learning concepts, particularly in natural language processing (NLP), large language models (LLMs), and generative AI. I focus on hosting, messaging, memory, and context. I demonstrate analytical skills to address complex technical challenges and devise innovative solutions.

Technical Skills:

- **Azure AI Components** – Azure AI Search, ML Studio, Language Services, Azure Form Recogniser
- **DevOps & Tools:** Containerisation, Docker, Kubernetes, Azure DevOps, Azure Monitor, PowerShell, YAML, Git, GitHub Actions, Terraform, SCA Tools, ARM Templates, Bash Scripting, Networking Basics, Build and Test automation Integrations
- **Cloud Platforms:** Microsoft Azure (Azure PaaS, App Service – WebApp, Container App, Function App, Logic App, APIM, ADF, Cosmos DB, Key Vault), GCP (Cloud Run, API Gateway, API Gee, BigQuery), Azure Kubernetes Services and Docker
- **Programming:** Python, C#, SQL, TypeScript
- **Frameworks:** .NET Core, ASP.NET MVC/Web API, Entity Framework, Flask, SQL Alchemy, FatAPI, Flask
- **Monitoring:** Prometheus, Grafana, Azure App Insights, Azure Log Analytics, Azure Monitor
- **Databases:** Azure SQL, SQL Server, Data Bricks, Cosmos DB, Google BigQuery, Chroma, PG Vector, SQLite
- **Data Analytics Platform and Data Pipelines:** Microsoft Fabric, Azure Data Bricks, Azure Data Lake, Azure Data Factory
- **Authentication:** OAuth2, Azure Entra ID, Okta, AWS Cognito, OpenID Connect
- **Architecture:** Micro services, Event-Driven, Server-less, RESTful APIs, DevOps Architecture, API Gateway, Enterprise Messaging (Azure Service Bus, Rabbit MQ), Data Streaming
- **Deployment Patterns** – Micro Service, Blue/Green, Rolling Deployment
- **AI Centric Stack:** LangChain, PromptFlow, MCP, A2A, RPC for AI Chat bot, Streamlit, ACP
- **Frontend Stack** – React, Vite, NextJS, Vitest, Jest, Typescript, Playwright

Certifications

- Microsoft Certified: Azure AI Fundamentals (AI-900)
- Microsoft Certified: Azure Fundamentals (AZ-900)
- Microsoft Certified: Azure Solutions Architect Expert (AZ-305)
- Microsoft Certified: Azure Architect Technologies (AZ-303)
- Microsoft Certified: Azure Developer Associate (AZ-204)

Professional Experience

Cognizant– Architect May 2019 – Present | Kolkata, India

Solution from inception to hypercare, Led Architecture design, development, Integration, Data Engineering, DevOps, Architecture of AI/ML-powered solutions using Azure OpenAI, FastAPI, Python, Azure Data Bricks, Data Ops and Data Engineering.

Architecture Governance & Oversight: Lead the end-to-end architecture governance process for AI/ML and cloud-native solutions, ensuring all designs adhere to enterprise scalability, performance, and security standards.

Stakeholder Alignment: Propose and defend architectural designs to Technical/Domain Architects, Enterprise Architects, and FinOps boards to align technical execution with organizational business goals.

Strategic Design & Reuse: Define and maintain shared architectural blueprints and reusable design patterns for data acquisition and GenAI modules, reducing time-to-market and increasing consistency across business units.

AI/ML Integration Excellence: Architected conversational AI products and Generative AI modules using Python, FastAPI, and Azure OpenAI, integrating RAG, prompt flows, and vector databases (Chroma, PG Vector).

DevOps & Security Standards: Spearheaded the integration of automated security assessments—including SAST, SCA, and container scanning—into CI/CD workflows to enforce compliance in regulated environments.

Cloud Infrastructure Governance: Manage GitOps, DevOps and Infrastructure as Code (Terraform, YAML) to ensure the continuous reconciliation of cloud infrastructure with established security and operational policies.

Data Platform and Analytics: With my current assignment, I am transforming data platform for a Manufacturing Enterprise, with Azure and Microsoft Fabric



Assignment: Generative AI Solution in Pharma Supply chain domain

Oct 2023 – Mar 2026

Role: Technical Architect | Tech Lead | Tech Stack: Python, FastAPI, LangChain, Streamlit, Azure OpenAI, Azure SQL, Azure Storage, Azure Data Bricks, React, Docker, KONG API Gateway, Azure PaaS, Azure DevOps

- Experience leading the development of multiple GenAI modules, including prompt flow, Conversational AI product development, data lake and integration with Python, PySpark, ADLS, and Azure Data Bricks.
- Expertise in leading the architecture and development of generative AI applications, integrating authentication and authorisation using Azure AD and OAuth2 for compliance with healthcare data regulations.
- Experience collaborating with cross-functional teams to align AI solutions with business goals and regulatory standards.
- Skilled in delivering reusable architectural patterns and CI/CD pipelines for rapid deployment and scaling.
- Proficient in integrating Code Scan, Container Scan, SAST, SCA, and Dependency Scan with CI/CD workflows.
- Familiarity with SonarQube, Dependency Scan, GitHub Advanced Security (Dependabot, CodeQL, Secret Scan).
- Capable of designing an AI-driven document automation platform for compliance and governance.
- Proficient in developing REST APIs, WebSockets, RPC, and leading the cloud deployment strategy for integrating an end-to-end solution.
- Committed to assisting the team and organisation in adopting AI-assisted code generation with GitHub CoPilot.



Assignment: Partner Compensation for Logistics Domain (May-2022 – Sep-2023)

Role: Technical Architect | **Tech Lead** | **Tech:** .Net, WebAPI, Docker, GCP BigQuery, GCP Storage, Docker, Azure DevOps, Team handling, Solution Structuring and Project Execution

- Collaborated with clients from inception to hyper care, focussing on cloud and application architecture, API development, and CI/CD.
- Led the technical development of multiple modules and integrated end-to-end solutions.
- Provided reusable architectural patterns and CI/CD pipelines for rapid deployment and scaling.
- Integrated code scan, container scan, SAST, SCA, and dependency scan with CI/CD workflows.



Assignment: Retail Fast Service Restaurant – Inventory Management/ Back of House (Jun-2019 – Apr-2022)

Role: Tech Lead | **Tech:** .Net, WebAPI, Docker, Azure DevOps, Azure Storage, Docker, Azure SQL, Azure API Management, C#, Mobile App Integration, Team handling, Solution Structuring, Technical Architecture

FNF India Pvt Ltd – Technical Lead / Senior Software Engineer May 2016 – Apr 2019 | Bangalore, India

- Developed enterprise-solutions .NET applications for the US title insurance domain.
- Led backend API development and integration with Angular frontends, Full text search (Elastic search), and a relational

database in a private cloud infrastructure.

- Mentored junior developers and contributed to Agile planning and delivery.

Infinite Computer Solutions – Senior Software Engineer May 2015 – Apr 2016 | Bangalore, India

Developed and Lead junior developers to develop enterprise-solutions .NET MVC Application stack, Entity Framework and PowerShell Scripts

Nexval Infotech – Software Developer Oct 2014 – May 2015 | Kolkata, India

Developed enterprise-solutions .Net, C#, ASP.NET MVC Application stack, Entity Framework.

National Informatics Centre – Software Developer / Support Personnel Dec 2008 – Sep 2014 | Kolkata, India

Developed enterprise-solutions and web applications with VB6, VB.Net, .Net, C#, ASP.NET Application stack

Education:

- Masters in Computer Application (MCA), West Bengal University of Technology – 2008

- B.Sc. (Math, Physics, Chemistry), University of North Bengal – 2005